

Transsplenic Portal System Catheterization: Review of Current Indications and Techniques

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Abbreviations: DSA = digital subtraction angiography, NBCA = N-butyl cyanoacrylate, PV = portal vein, PVE = PV embolization, PVR = PV recanalization, PVS = PV stenosis, PVT = portal vein thrombosis, SV = splenic vein, TIPS = transjugular intrahepatic portosystemic shunt, TSA = transsplenic access

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SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Discuss the clinical indications for accessing the portal system through percutaneous TSA.
- Describe the steps of percutaneous portal system TSA and postintervention management of access.
- Identify the common complications associated with percutaneous TSA.

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Multiple diseases of the portal system require effective portal vein access for endovascular management. While percutaneous transhepatic and transjugular approaches remain the standard methods of portal vein access, transsplenic access (TSA) has gained recognition as an effective and safe technique to access the portal system in patients with contraindications to traditional approaches. Recently, the utility of percutaneous TSA has grown, with described treatments including recanalization of chronic portal vein occlusion, placement of stents for portal vein stenosis, portal vein embolization of the liver, embolization of gastric varices, placement of complicated transjugular intrahepatic portosystemic shunts, and interventions after liver transplant. The authors provide a review of percutaneous TSA, including indications, a summary of related portal vein diseases, and the different techniques used for access and closure. In addition, an imaging-based review of technical considerations of TSA interventions is presented, with a review of potential procedural complications. With technical success rates that mirror or rival the standard methods and reported low rates of major complications, TSA can be a safe and effective option in clinical scenarios where traditional approaches are not feasible.

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Introduction

Percutaneous transsplenic access (TSA) was first used in 1951 for splenoportography but was temporarily abandoned because of an increased incidence of hemorrhagic complications (1). The first TSA intervention was described by Tsang et al (2) in 1985 when the approach was used to correct stenosis in a splenorenal shunt. Since then, TSA has been selectively used in the endovascular management of multiple diseases of the portal system, particularly those related to sequelae of portal hypertension or malignancy. Specific examples include portal vein (PV) recanalization (PVR) after chronic obliterative occlusion, placement of stents for PV stenosis (PVS) after a transplant or because of malignancy, PV embolization (PVE) for hepatic hypertrophy, embolization of bleeding gastric varices, placement of a transjugular intrahepatic portosystemic shunt (TIPS), and after liver transplant (3–11).

Access to the portal system has been traditionally accomplished by means of a percutaneous transhepatic approach or by crossing from the hepatic vein to the PV through a transjugular approach, as in placement of a TIPS (12–14). However, several physiologic and anatomic conditions may limit the feasibility of these two options in endovascular interventions, resulting in the need for safe alternative routes of catheterization such as TSA. The purpose of this article is to review PV diseases amenable to intervention from a TSA approach and the indications and contraindications for TSA. This article also

TEACHING POINTS

- Several physiologic and anatomic conditions may prevent effective endovascular access through a transhepatic or transjugular route. Risk of failure of a transhepatic or transjugular route may increase if no patent intrahepatic venous or PV branches exist (related to chronic occlusion or stenosis) or if the main PV is occluded.
- In patients who are being considered for percutaneous TSA catheterization, preprocedural cross-sectional contrast-enhanced imaging is recommended to delineate the native anatomy and patency of the portal system, including the splenic vein (SV)–superior mesenteric vein confluence, the PV, and associated branches.
- PVR has traditionally been performed with a transhepatic approach; however, in a recent case series of 61 patients with cirrhosis and chronic PVT, a TSA approach used in 20 patients was described as technically easier, with a success rate of 100%.
- Although stent placement for PVS has also traditionally been performed with a transhepatic approach, a TSA technique has been increasingly used in select patients. The benefits are most notable in patients who have recently undergone a liver transplant, in whom TSA allows direct injury to the graft to be avoided.
- For clinical scenarios in which a traditional transhepatic approach is not feasible in patients who require PV intervention, TSA has been shown to be a generally safe technique with high technical success rates.

provides a comprehensive imaging-based review of technical considerations of TSA catheterization, a summary of described access and tract embolization techniques in the literature, and a discussion regarding reported procedural complication rates.

Access

Several physiologic and anatomic conditions may prevent effective endovascular access through a transhepatic or transjugular route. Risk of failure of a transhepatic or transjugular route may increase if no patent intrahepatic venous or PV branches exist (related to chronic occlusion or stenosis) or if the main PV is occluded (8). Additional scenarios include diffusely infiltrating malignancy in the trajectory of percutaneous access (ie, risk of tumor seeding), or perihepatic fluid (ie, ascites or hematoma) (6,8,15–18). In addition, in patients undergoing liver transplant, avoidance of direct injury to the allograft from a percutaneous approach may warrant consideration of a TSA intervention. Before the procedure, it is pivotal to be aware of the patient's preexisting conditions and anatomy. The basic anatomy of the hepatic venous and portal systems is shown in Figure 1 (19).

Preprocedural Assessment

In patients who are being considered for percutaneous TSA catheterization, preprocedural

cross-sectional contrast-enhanced imaging is recommended to delineate the native anatomy and patency of the portal system, including the splenic vein (SV)–superior mesenteric vein confluence, the PV, and associated branches (3,5,6,8,9,11,20,21). The presence of pathologic varices or coexisting neoplastic processes should also be assessed preprocedurally. The feasibility of TSA to the portal system is decided by the operator on the basis of anatomic assessment, including the presence of vessel tortuosity, diameter, and patency (21).

Percutaneous TSA

In existing studies (5,6,8–11,16,20,21) performed to assess the use of TSA, several common technical steps are described. The patient is placed in the supine position or with the left side of the body slightly raised, and the spleen is assessed with gray-scale and color Doppler US. The use of general anesthesia with paralytic medication is recommended; the ability to suspend respiration allows optimal assessment of the spleen and TSA. Under real-time US guidance, a central intraparenchymal perihilar branch of the SV is identified and accessed percutaneously with a 21–22-gauge Chiba needle (Cook Medical), from a lateral intercostal or subcostal approach. Care should be taken to access the vessel in a trajectory that facilitates in-line access of the central SV with a wire and vascular sheath, rather than approaching the vessel at an angle that is too acute. Authors of at least two studies (9,21) have described injecting a small amount of iodinated contrast material under fluoroscopic visualization to confirm SV branch access. Parenchymal injection of contrast material may show opacification of the SV; gentle advancement of an 0.018-inch guidewire can allow confirmation of adequate access.

After successful puncture, the needle is exchanged over an 0.018-inch guidewire to accommodate a 5F catheter. This is commonly performed with an introducer set (Accustick; Boston Scientific), with upsizing to a 35–45-cm 5F sheath (8–10, 20). Ohm et al (11) and Ko et al (21) also described an alternative initial access with the use of a 6F coaxial dilator (Neff sheath; Cook Medical) and a 6F–7F sheath (Radifocus Introducer II sheath; Terumon). Portal venography can then be performed as needed, with advancement of a 5F catheter of the operator's choice over an 0.035-inch guidewire for subsequent endovascular intervention. Venography often demonstrates patent native intrahepatic portal vessels secondary to developed collateral flow. Examples of this process are provided in Figure 2. Additional images of the trajectory of needle access are shown in Figures 3B, 4B, and 5A.

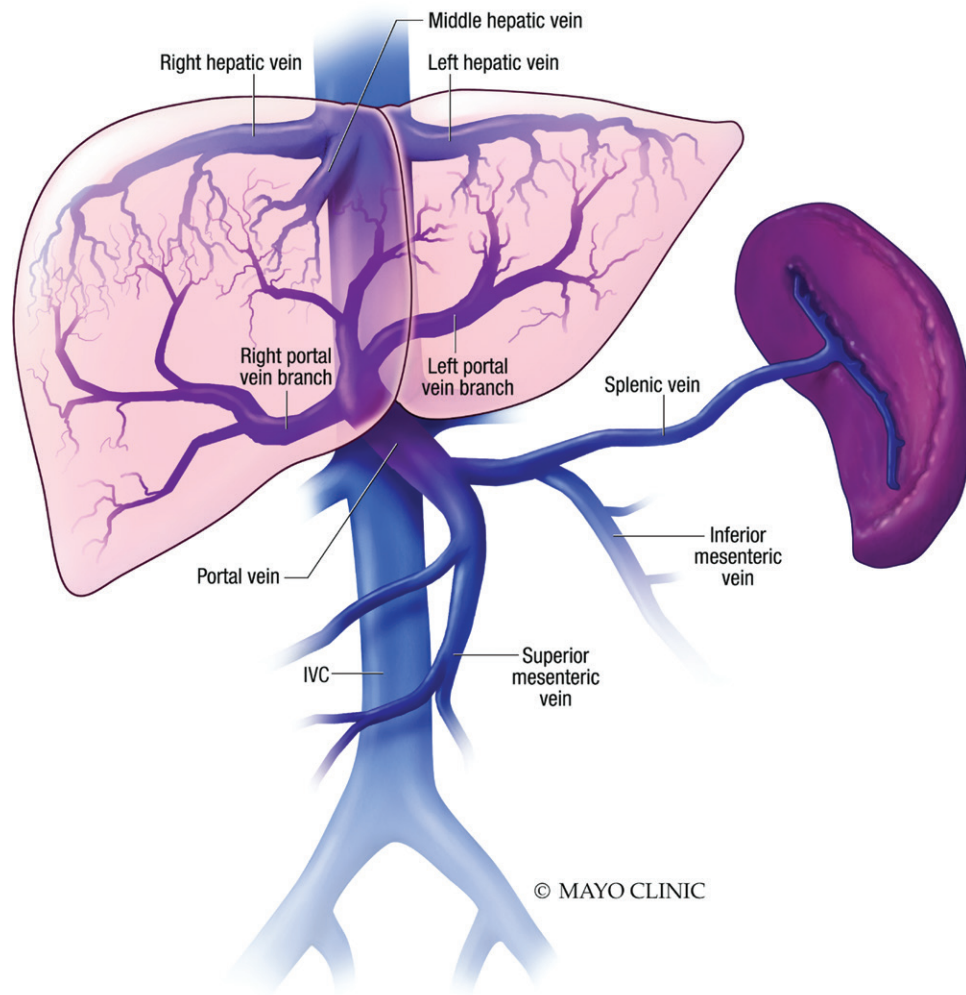


Figure 1. Illustration shows the hepatic and portal vein (PV) systems and their relationships to the liver and spleen. IVC = inferior vena cava. (Reprinted, with permission from Mayo Foundation of Medical Education and Research, from reference 19.)

Interventions

PV Recanalization for Chronic PV Thrombosis

Chronic PV thrombosis (PVT) is a common complication of portal hypertension in patients with cirrhosis. Patients may present with common sequelae of portal hypertension, such as variceal bleeding and ascites (22,23). In some medical centers, PVT remains an absolute or relative contraindication to liver transplant; analysis of the United Network for Organ Sharing registry has shown PVT to be an independent risk factor for posttransplant death (24). In patients undergoing a transplant, characterizing the extent of the thrombus is essential for procedural planning. The most widely used classification system was described by Yerdel et al (25) in 2000. The Yerdel classification system allows grading of PVT on the basis of involvement of the PV and the superior mesenteric vein. In grade 1 PVT, less than 50% of the PV lumen is involved; in grade 2 PVT, greater than 50% of the lumen is involved; grade 3 is complete PVT with extension into the

proximal superior mesenteric vein; and Grade 4 is complete PVT with occlusion of both the proximal and distal superior mesenteric vein. Yerdel et al (25) found that patients with grade 1 PVT who underwent liver transplant had outcomes as favorable as those of control subjects, with grades 2–4 having increasingly poorer outcomes and patients of all grades requiring more technically difficult surgical procedures (25,26). PVR with placement of a TIPS can be performed in select cases of PVT, thereby allowing select patients to regain candidacy to undergo liver transplant (3,20). PVR has traditionally been performed with a transhepatic approach; however, in a recent case series (9) of 61 patients with cirrhosis and chronic PVT, a TSA approach used in 20 patients was described as technically easier, with a success rate of 100%.

Summaries of techniques for PVR and placement of TIPS have been provided in several existing articles (9,20). Both Thornburg et al (9) and Habib et al (20) describe initial internal jugular vein access, with confirmation of the presence of PVT reconfirmed with retrograde

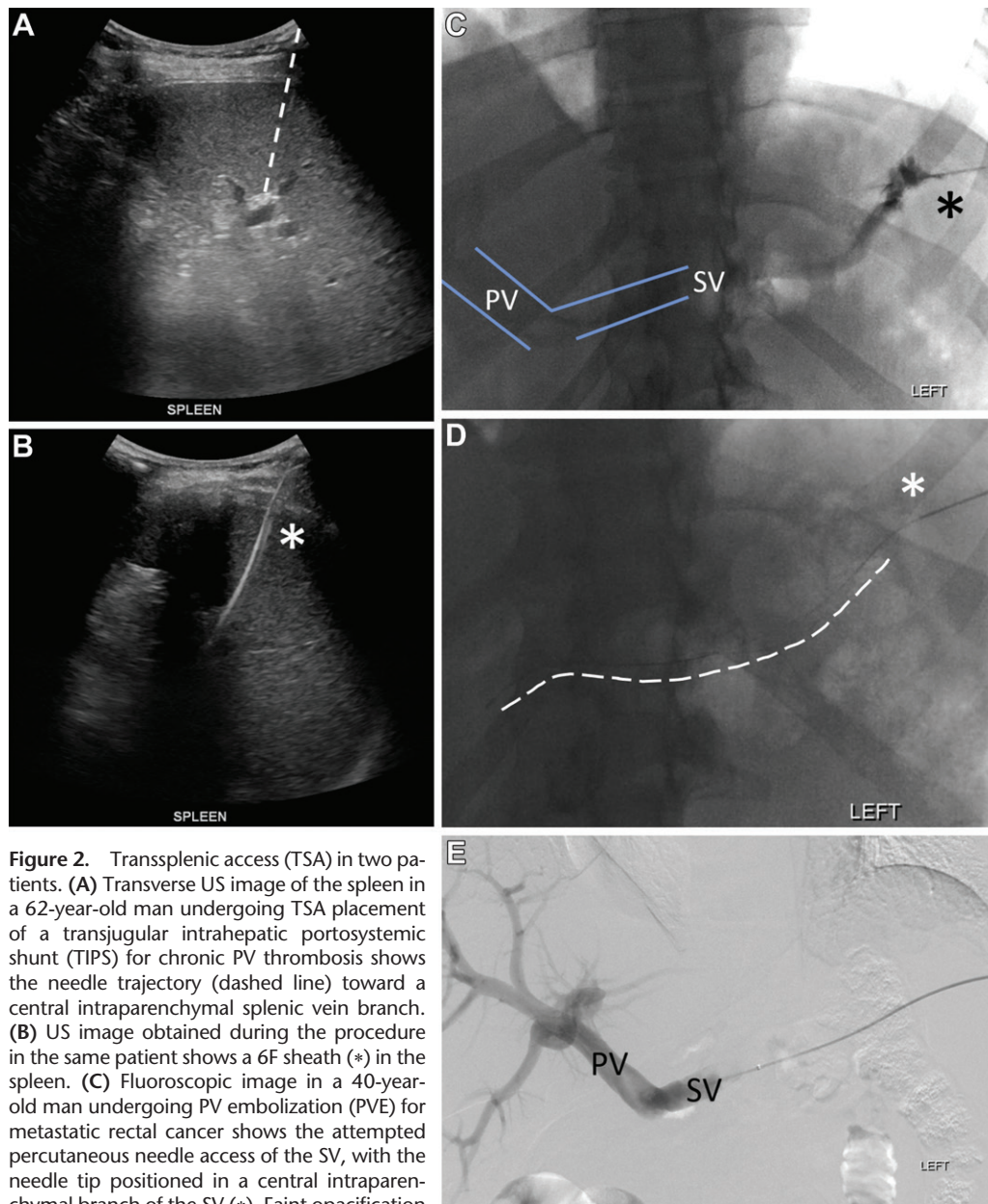


Figure 2. Transsplenic access (TSA) in two patients. (A) Transverse US image of the spleen in a 62-year-old man undergoing TSA placement of a transjugular intrahepatic portosystemic shunt (TIPS) for chronic PV thrombosis shows the needle trajectory (dashed line) toward a central intraparenchymal splenic vein branch. (B) US image obtained during the procedure in the same patient shows a 6F sheath (*) in the spleen. (C) Fluoroscopic image in a 40-year-old man undergoing PV embolization (PVE) for metastatic rectal cancer shows the attempted percutaneous needle access of the SV, with the needle tip positioned in a central intraparenchymal branch of the SV (*). Faint opacification of the SV and PV (blue lines) appears after injection of contrast material, confirming SV access. (D) Fluoroscopic image in the same patient as in C shows an 0.018-inch wire advancing through the needle at the percutaneous access site (*) into the SV (dashed line). (E) Portogram after placement of a 6F sheath shows that portography demonstrates optimal anatomic assessment of the SV and PV.

wedged portography. At our institution, splenic access is acquired first, with the remaining procedural techniques identical to those described in these studies. In addition, as described by Thornburg et al (9) and in our experience, the thrombosed PV can often be identified at the superior aspect of the PV confluence, where a thin residual lumen is often present. The PV should course superolaterally; if the left gastric vein is opacified on a PV venogram, this may also be used as a landmark for the expected location of the PV origin. PVR is performed with the patient under general anesthesia. US-guided TSA is then performed. An example of

PVR and placement of a TIPS is presented in Figure 6.

After the procedure, care should be taken to minimize both the length of the wire uncovered by venous and PV sheaths and wire tension to prevent laceration of the liver parenchyma (9,20). Completion venography can be performed, and pressure can be measured to confirm the decompression of the portal system. TSA embolization can be performed with various materials. Use of postprocedural anticoagulation medication has not been routinely described. Patients can return in 1 month for surveillance venography or screening US of the TIPS, as needed.

Figure 3. TSA for right PVE in a 70-year-old man. (A) Axial delayed gadolinium-enhanced T1-weighted MR image shows hepatocellular carcinoma in the right hepatic lobe (arrow). (B) US image shows US-guided TSA of an anterior branch of the SV, with a 21-gauge Chiba needle (dashed line). (C) DSA image after placement of a 5F catheter through a 6F 25-cm sheath shows the right and left PV branches. (D) Repeat portogram acquired after right PVE with 300–500- μ m and 700–900- μ m microspheres (Embospheres; Merit Medical Systems) and coils (Azur; Terumo Medical) shows non-opacification of the right PV branches, confirming successful embolization.

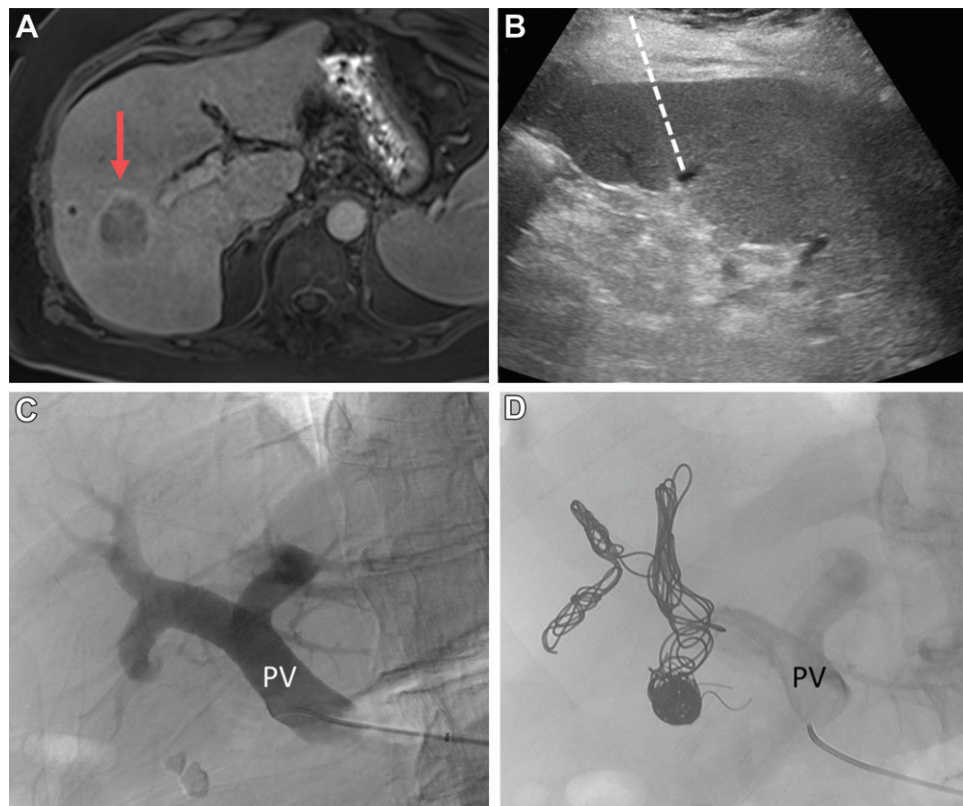


Figure 4. TSA embolization and obliteration of bleeding gastric varices in a 65-year-old man with chronic PV occlusion and gastric varices. (A) Axial T1-weighted contrast-enhanced subtraction MR image shows prominent mucosal gastric varices (arrow). (B) US image shows access to the short gastric vein with a 20-gauge Chiba needle (dashed line). (C) Digital subtraction angiographic (DSA) image after advancement of a 6F sheath and a 5F catheter shows known draining gastric varices superiorly (*). SGV = short gastric vein. (D) After exchange of the 5F catheter for a Foley balloon catheter, a 1:1:1 solution of sodium tetradecyl, ethiodized oil, and air was injected. Repeat DSA image of the SV after embolization of the gastric varices shows an 8-mm vascular plug (arrow) and residual ethiodized oil in the embolized gastric varices (*).

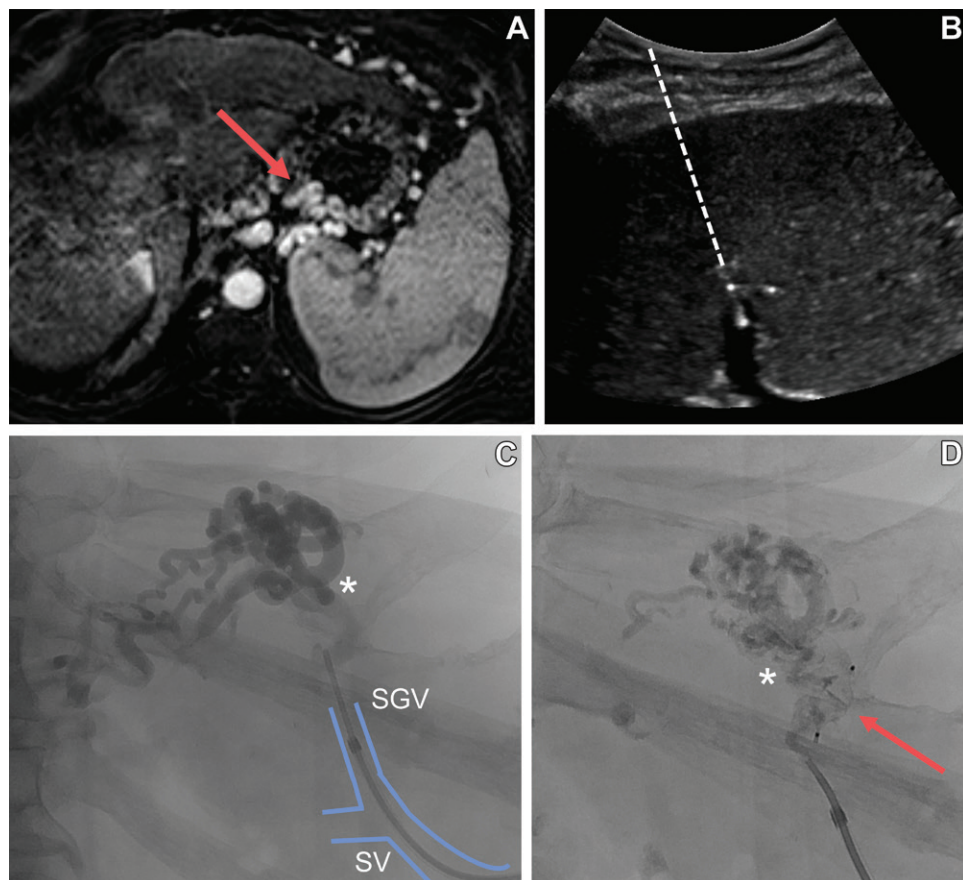
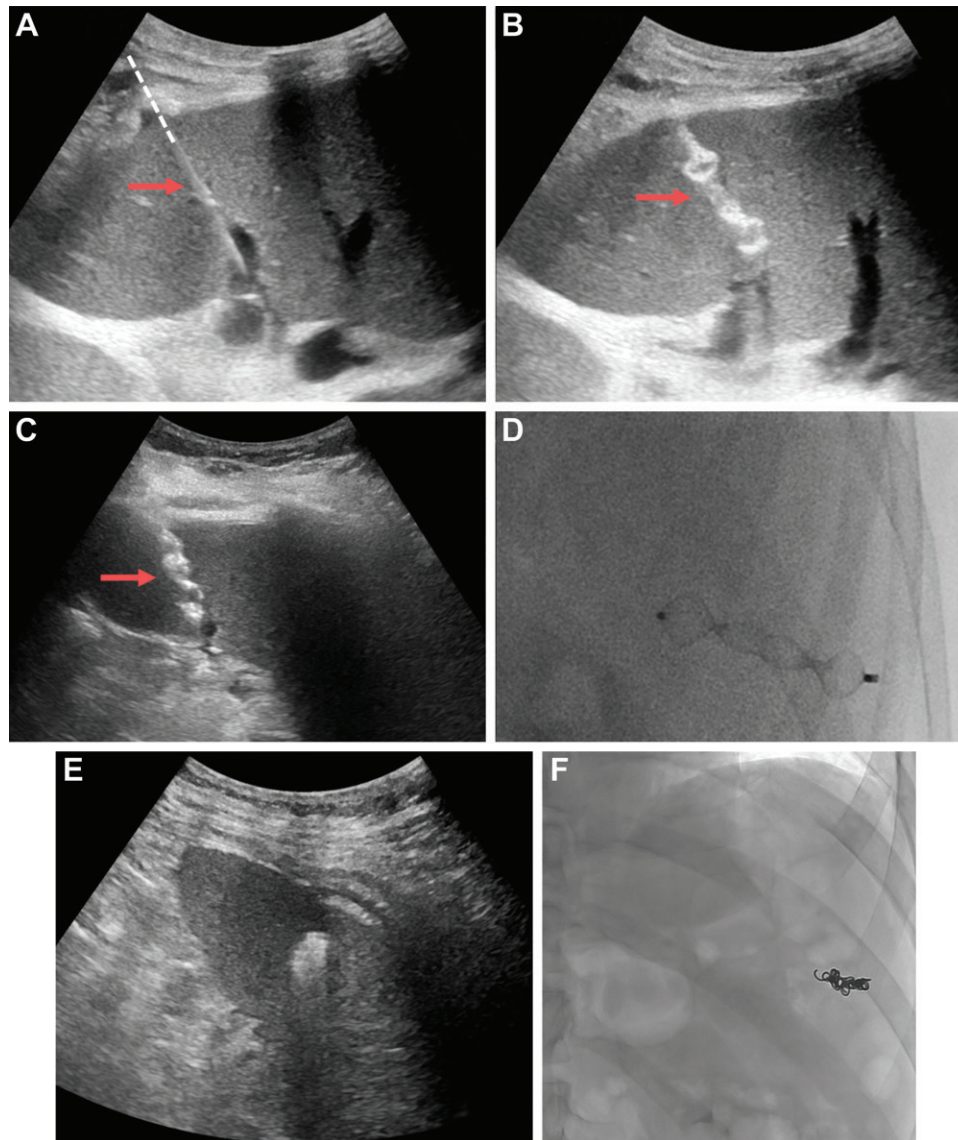


Figure 5. Examples of TSA and tract embolization. (A) Transverse US image shows percutaneous 21-gauge needle access of an anterior branch of the SV (arrow, dashed line). (B, C) Transverse US images show tract embolization in two patients with a vascular plug (Amplatzer Vascular Plug II), with the edge of the plug extending to the splenic capsule (arrow). (D) Digital fluoroscopic image shows the same type of vascular plug as that in B and C. (E, F) Transverse US image (E) and fluoroscopic image (F) show tract embolization with coils.



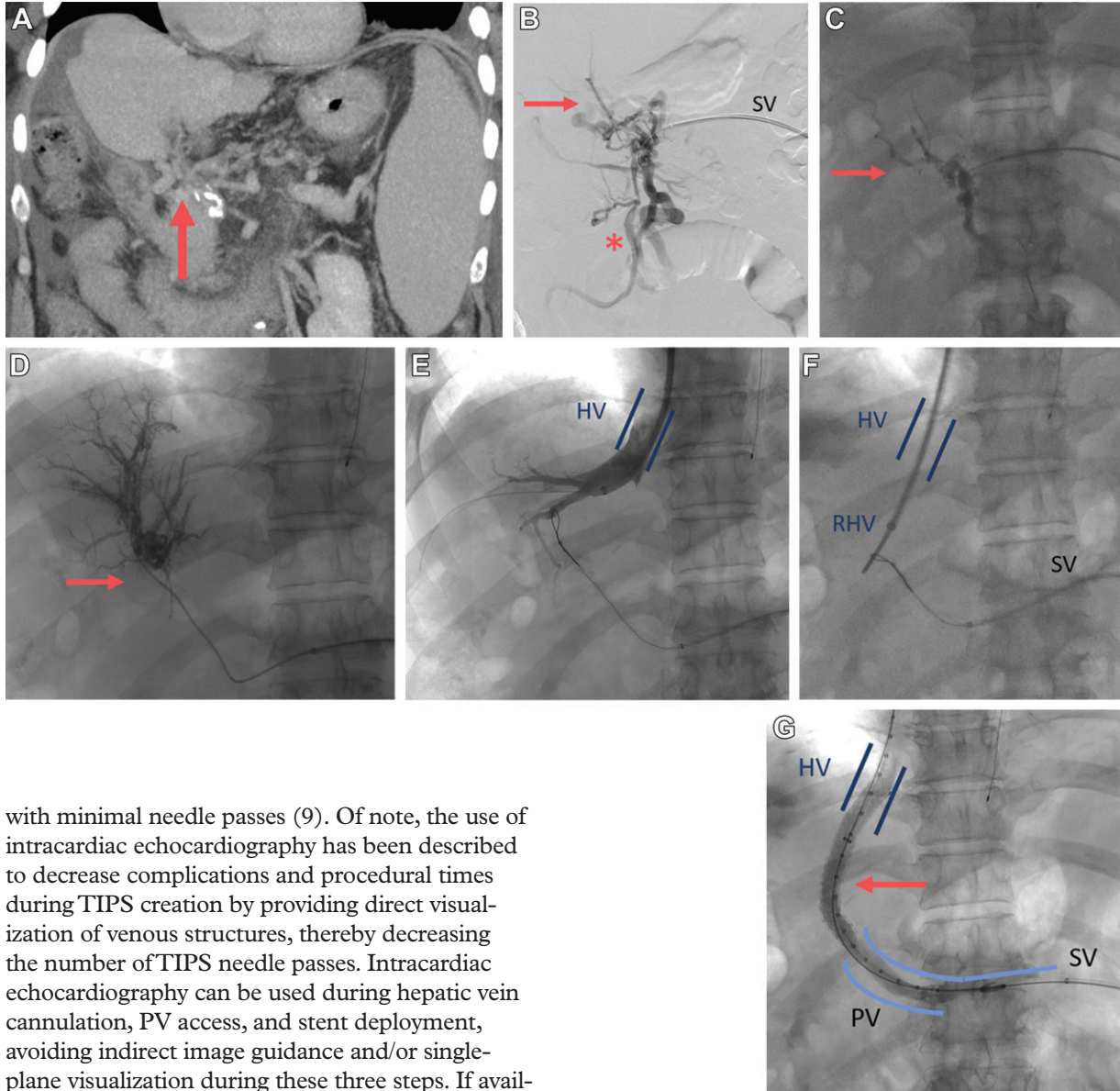
Complex TIPS Placement Procedures and Stent Dysfunction

The TIPS placement procedure is an effective form of PV decompression and is typically used to manage the sequelae of portal hypertension. The primary indications for placement of a TIPS that have the strongest evidence include secondary prevention of esophageal variceal bleeding and treatment of refractory ascites (27). Additional indications include hepatorenal and hepatopulmonary syndrome, Budd-Chiari syndrome, or treatment of acute variceal bleeding.

Although it is typically performed with a transhepatic transvenous approach, placement of a TIPS can also be performed using a TSA, if no suitable

hepatic veins are available. Similarly, in patients with Budd-Chiari syndrome or TIPS occlusion, recanalization after stent dysfunction with a TSA is also possible, although to our knowledge, no large-scale retrospective or prospective trials have been performed to assess this technique (7,27). TSA is also helpful in patients who are unable to tolerate the liver trauma of several passes with sharp transhepatic canalization, particularly in those with cirrhosis. In a technique similar to that of PVR, positioning of a snare into the portal system provides a fluoroscopic target when the TIPS needle is passed through the liver parenchyma during placement. Slow incremental advancement of the needle under multiple fluoroscopic obliquities allows for snaring

Figure 6. Transsplenic recanalization of chronic obliterative PV obstruction in a 64-year-old woman with cirrhosis and a chronic obliterative PV occlusion who was awaiting a liver transplant. **(A)** Coronal contrast-enhanced CT image shows chronic PV occlusion and cavernous transformation, with a calcified component (arrow). **(B)** Venogram shows the known PV occlusion, with the collateral veins of the PV (arrow) and the superior mesenteric vein (*) after TSA was acquired with a 21-gauge needle and a 6F sheath. **(C, D)** Portograms show that a right PV branch was successfully accessed (arrow) with an angled-tip 5F catheter (Kumpe; Cook Medical) and an 0.035-inch guidewire (GlideWire; Terumo). A gooseneck snare (Medtronic) was placed in the right PV branch in exchange for the catheter. **(E)** Hepatic venogram shows that the hepatic vein (HV) access was obtained from a right internal jugular vein approach. **(F)** Fluoroscopic image shows targeting of a right PV snare by a right hepatic vein TIPS needle. With fluoroscopic guidance at multiple obliquities and following the Seldinger technique, a 10F TIPS needle was advanced through a right hepatic vein (RHV) branch into the right PV toward the gooseneck snare. Once the needle was grasped, a guidewire was snared through the portovenous access, and after angioplasty, an 8–10-mm $\times$ 7-cm TIPS (Viatorr; Gore) was placed. HV = hepatic vein. **(G)** Portogram shows the patent TIPS (arrow) extending to the inferior vena cava. The patient underwent a liver transplant 1 month later. HV = hepatic vein.



with minimal needle passes (9). Of note, the use of intracardiac echocardiography has been described to decrease complications and procedural times during TIPS creation by providing direct visualization of venous structures, thereby decreasing the number of TIPS needle passes. Intracardiac echocardiography can be used during hepatic vein cannulation, PV access, and stent deployment, avoiding indirect image guidance and/or single-plane visualization during these three steps. If available, intracardiac echocardiography could be used in place of the TSA snare technique previously described in this section to avoid the trauma of several passes as the TIPS needle is advanced through the liver parenchyma. An example of placement of a TIPS in a patient with attenuated hepatic veins is provided in Figure 7.

PV Stenosis

PVS can occur because of multiple clinical entities and accounts for 5%–10% of all cases of

portal hypertension (28). In severe cases, patients may present with clinical sequelae of portal hypertension. Commonly described causes include postoperative stenosis after liver transplant, postinflammatory changes related to pancreatitis, and mass effect or direct invasion related to oncologic processes (29–31). Fifteen to twenty-four percent of cases of extrahepatic PV stenosis are the result of hepatobiliary and/or pancreatic

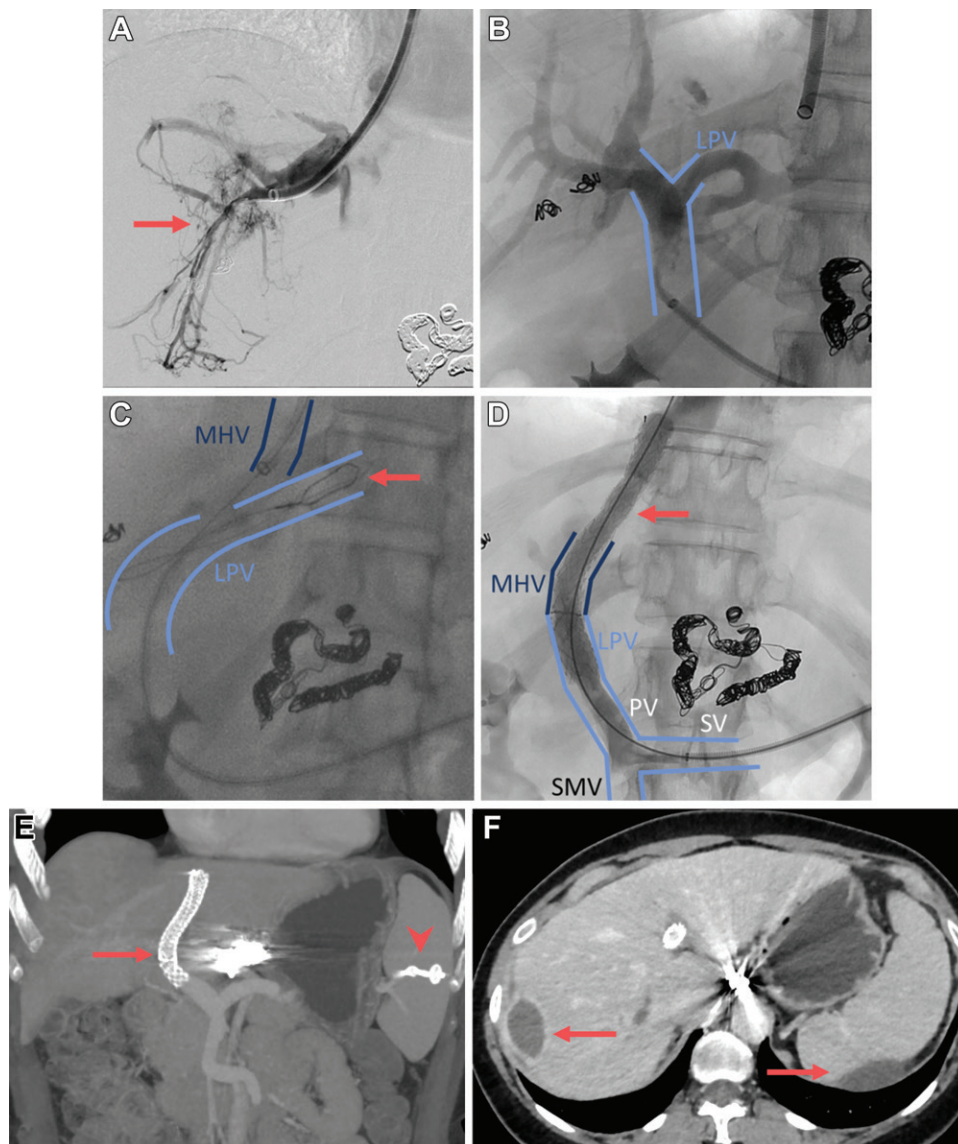


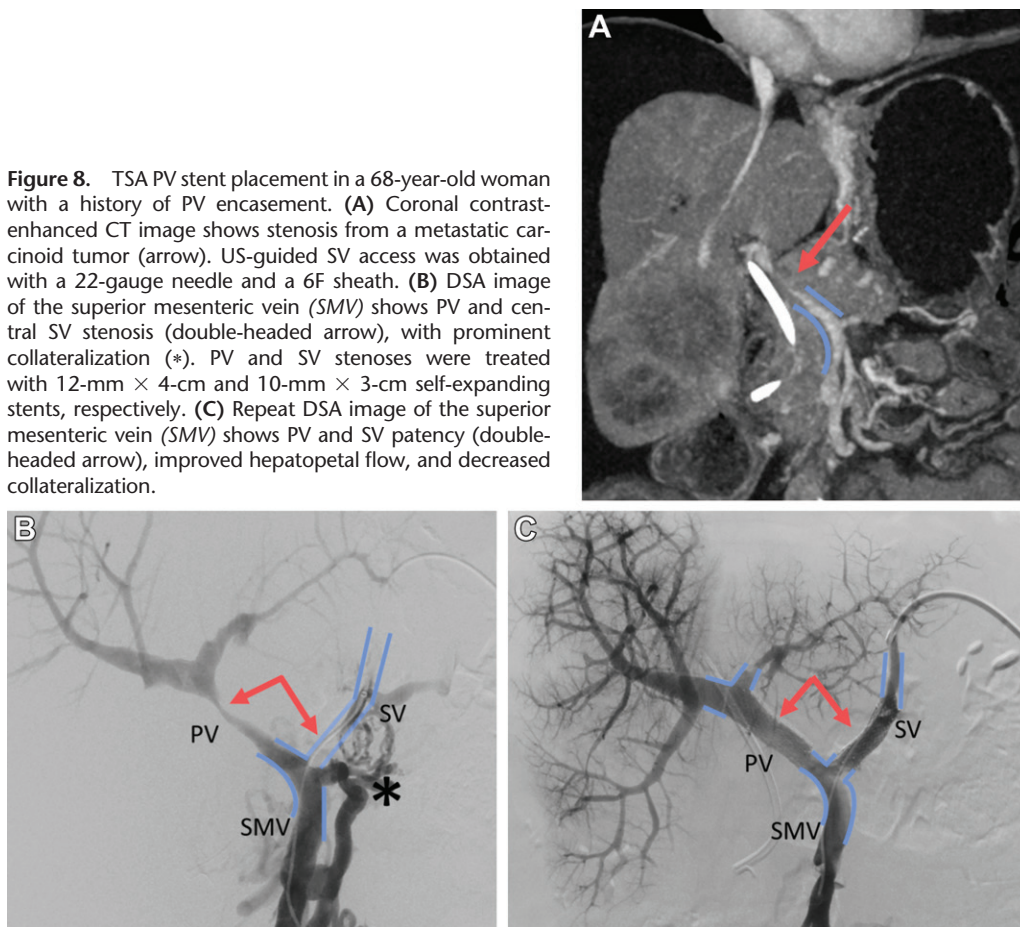
Figure 7. Combined TSA approach to placement of a TIPS in a 66-year-old woman with portal hypertension and gastroesophageal varices. (A) DSA image shows that initial attempts at TIPS placement were unsuccessful because of right hepatic vein stenosis (arrow). To facilitate a middle hepatic vein to left PV approach, US-guided SV access was obtained. (B) DSA image shows a patent left PV (LPV). (C) Fluoroscopic image shows a gooseneck snare (Medtronic) as it is advanced into the left PV (LPV) through a 5F Kumpe catheter and a 45-cm 6F sheath. A TIPS needle was used to cross from the middle hepatic vein (MHV) to the left PV, through which a guidewire (GlideWire; Terumo) was advanced and ensnared (arrow). (D) DSA image shows successful deployment of a stent (Viatorr; Gore) after portosystemic access was obtained (arrow). LPV = left PV, MHV = middle hepatic vein, SMV = superior mesenteric vein. (E) Coronal contrast-enhanced maximum intensity projection CT image acquired at 1-month follow-up shows the patent TIPS extending from the PV to the hepatic vein (arrow) and a partially visualized vascular plug in the spleen (arrowhead). (F) Axial contrast-enhanced maximum intensity projection CT image shows small resolving perisplenic and perihepatic hematomas (arrows), presumably related to TSA and TIPS access, respectively. No intervention was required.

malignancies (31,32). PVS is uncommon in those undergoing a liver transplant, occurring in 1%–2% of patients receiving whole liver grafts, almost always at the anastomosis (33).

Endovascular intervention has become the mainstay of treatment in patients with PVS in the settings of malignancy and after transplant, with multiple studies (28,31,34–36) demonstrating that placement of a transhepatic PV stent is safe and

effective and that these stents have good long-term patency of 60%–100% at 12–66 months. Although stent placement for PVS has also traditionally been performed with a transhepatic approach, a TSA technique has been increasingly used in select patients. The benefits are most notable in patients who have recently undergone a liver transplant, in whom TSA allows direct injury to the graft to be avoided (11). Furthermore, in a patient with PVS,

Figure 8. TSA PV stent placement in a 68-year-old woman with a history of PV encasement. **(A)** Coronal contrast-enhanced CT image shows stenosis from a metastatic carcinoid tumor (arrow). US-guided SV access was obtained with a 22-gauge needle and a 6F sheath. **(B)** DSA image of the superior mesenteric vein (SMV) shows PV and central SV stenosis (double-headed arrow), with prominent collateralization (*). PV and SV stenoses were treated with 12-mm $\times$ 4-cm and 10-mm $\times$ 3-cm self-expanding stents, respectively. **(C)** Repeat DSA image of the superior mesenteric vein (SMV) shows PV and SV patency (double-headed arrow), improved hepatopetal flow, and decreased collateralization.



intrahepatic PV branches may collapse, increasing the technical difficulty of percutaneous transhepatic access. If PVS is related to hepatobiliary malignancy, the tumor burden in the trajectory of access may also limit traditional access. An example of placement of a PV stent appears in Figure 8.

In two studies, Ohm et al (11) and Zhu et al (8) assessed the safety and efficacy of TSA in various PV procedures. In the study by Ohm et al (11), TSA was used to avoid graft injury in patients undergoing liver transplant. Six patients underwent PV stenting for PVS, with or without variceal embolization. Technical success was achieved in 100% of patients, with no major complications. In the study by Zhu et al (8), five patients received PV stents, with PVS related to malignant stenosis or a tumor thrombus. Technical success was achieved in 44 of 46 patients undergoing TSA (96%), with 100% patency at 4–12 months. The techniques used and the complication rates are further detailed in Tables 1 and 2.

PV Embolization

After a regional injury occurs, the liver can regenerate through hyperplasia at a rate proportional to the degree of injury (37). Regeneration can be seen after a toxic insult, vascular occlusion,

partial hepatectomy, or direct trauma, with up to 95% of noninjured cells undergoing mitosis after an insult involving greater than 10% of the liver (38). In patients with a primary liver neoplasm or secondary metastatic disease, surgical resection, embolization, ablation, or a liver transplant currently are the mainstays of temporizing or curative management. Although the potential for liver regeneration allows partial hepatectomy in select patients, large-volume resections place patients at risk for postoperative liver insufficiency (20,37,39). Before partial hepatectomy, it is essential to estimate the remaining volume of functioning liver to be retained; this residual tissue is termed the *future liver remnant* and can be calculated at cross-sectional imaging.

In patients whose future liver remnant is small enough to risk liver insufficiency, preoperative PVE of the diseased liver lobe or segments can improve future liver remnant volume by inducing compensatory hyperplasia of the unaffected lobe (39,40). In patients with normal liver function, a future liver remnant of greater than 20% has been associated with reduced postoperative complications, with a future liver remnant of less than 20% being considered a standard indication for PVE (37). Although patients with liver cirrhosis are

often considered to be poor candidates for surgery, in select patients with a Child-Pugh classification of A, a future liver remnant of less than 40% is considered to be an indication for PVE (41). Absolute contraindications include an extensive ipsilateral tumor (in which PV flow to the affected region is compromised, limiting the efficacy of embolization) and in portal hypertension (in which hepatectomy is contraindicated) (37).

PVE has traditionally been performed through a transhepatic approach, with transileocolic or transjugular techniques also described. For patients in whom an ipsilateral transhepatic approach is not accessible, or in those for whom prevention of direct injury to the future liver remnant is desired, a TSA approach has been shown to be both safe and feasible (21). In 2016 Ko et al (21) attempted 27 TSA PVEs in patients who could not undergo transhepatic PVE because of large or multiple tumors in the puncture trajectory, or a high-positioned liver. TSA PVE was successful in 89% of patients, with a single major complication (3.8%) related to SV dissection. Further detail of Ko et al's TSA technique is described in Table 1, with PVE being performed with a combination of gelatin plugs, N-butyl cyanoacrylate (NBCA) with ethiodized oil, coils, or vascular plugs (Amplatzer; Abbott). Embolization is performed primarily of first- and second-order PVs of the affected segments. Multiple embolic strategies for PVE have been described (39). Examples of TSA PVE performed at the authors' institution with coil embolization and NBCA-ethiodized oil embolization are provided in Figures 3 and 9.

Embolization of Varices

Chronic liver disease and portal system abnormalities can lead to gastrointestinal varices, which are dilated submucosal veins that have a potential to cause life-threatening bleeding (42). These varices develop secondary to portal hypertension to decompress the high-pressure portal system. Approximately 50% of patients with a new diagnosis of cirrhosis have gastrointestinal varices (43). The annual risk of variceal bleeding is approximately 5%–15%, and the 6-week mortality rate among patients with variceal bleeding is approximately 20%. Patient outcome depends greatly on stabilizing the patient and achieving hemostasis. Treatment options to help achieve hemostasis include lowering the portal system pressure, either pharmacologically with vasoactive medications or through placement of a TIPS, endoscopic tamponade, endoscopic ligation, sclerotherapy, embolization, or surgery (44). In patients with elevated PV pressure secondary to PV occlusion, thrombectomy or placement of a stent can help resolve bleeding (16).

Table 1: Reported Percutaneous TSA Techniques in Adults

Techniques and Supplies	References
Contrast-enhanced CT or MRI for preprocedural cross-sectional imaging assessment of the portal system	...
US intraprocedural assessment of spleen and SV branches	...
Percutaneous needle access	
US-guided access of the perihilar or intra-parenchymal branch of the SV	...
21-gauge needle	5,6,9–11,20,21
22-gauge needle	8,16
Fluoroscopic confirmation with small-volume contrast injection	9,37
Sheath placement	
Needle exchange over an 0.018-inch guidewire	...
Tract dilation	...
Introducer system	8–10,20*
Percutaneous access set	11,21†
Recommended needle gauge	
5F	8–10,20
6F–7F	11,21
4F–9F	6
Access tract embolization	
Vascular plugs	...
4-mm vascular plug	50‡
8-mm vascular plug	...
4-mm × 14-cm coils	9,20
Gelatin sponges and pledgets	10,16,49§
Glue: 1:2 mixture of NBCA: ethiodized oil	8
Combinations	
3-mm × 5-cm coils and NBCA	4 [†]
4-mm × 14-cm coils and 1:2 NBCA: ethiodized oil ± standard coils (various sizes)	6 [#]
Coils and 1:2 NBCA: ethiodized oil	11,21
Gelatin sponges and coils	5

Note.—NBCA = N-butyl cyanoacrylate.

* Authors of listed studies used the Accustick introducer system (Boston Scientific).

† Authors of listed studies used the Neff percutaneous access set (Cook Medical).

‡ Authors of the listed study used Amplatzer II vascular plugs (Abbott).

§ Authors of the listed studies used Surgifoam (Ethicon) or gelatin foam from Pharmacia and Upjohn.

[†] Authors of the listed studies used unspecified coils from Cook Medical.

[#] Authors of the listed study used Nester coils (Cook Medical).

Embolization and obliteration of bleeding gastroesophageal varices is typically performed either percutaneously through the transhepatic

Table 2: Reported Percutaneous TSA Complication Rates in Adults

Study*	Interventions	Sample Size	Sheath Size (F)	Bleeding		Other Complications	Tract Embolization
				Major	Minor		
Chu et al 2012 (6)	Variceal embolization, PVR, shunt closure	9	4–9	0	0	0	Coils and glue
Gong et al 2001 (5)	Variceal embolization	18	5	5 (28)	18 (100)	2 (11) [†]	Coils and gelatin sponge
Habib et al 2015 (20)	PVR and TIPS	11	5	0	0	2 (18) [‡]	Coils
Ko et al 2016 (21)	PVE	27	6–7	0	1 (4)	1 (4) [§]	Coils and glue
Liang et al 1997 (49)	Variceal assessment superselective PVE	...	4	2 (7)	3 (11)	0	Glue
Ohm et al 2017 (11)	Variceal embolization, balloon angioplasty, stent placement	10	6–7	1 (10)	0	1 (10) [§]	Coils and glue
Thornburg et al 2016 (9)	PVR and TIPS	61	5	2 (3)	0	0	Coils
Zhu et al 2013 (8)	Variceal embolization, PV stent placement, PVR, and TIPS	46	5	3 (7)	0	3 (7) [#]	Glue

Note.—Unless otherwise indicated, data are number of patients, with percentage in parentheses.

* Numbers in parentheses are reference numbers.

[†]Two related deaths.

[‡]One case of transient encephalopathy and one case of fever.

[§]One case of SV dissection.

^{||}Catheter only.

[#]Three cases of pleural effusion.

approach or through the transjugular approach with the placement of a TIPS. However, in cases where the intrahepatic PVs are occluded or attenuated, the risk of failure associated with these methods increases (8). As described in the literature (5,6,8,11,16), percutaneous TSA embolization of varices has been used in patients with PV and hepatic abnormalities that prevent the traditional approaches and has been proven to be a safe and effective method to treat esophageal and gastric variceal bleeding. Embolic materials used include coils, sodium tetradecyl sulfate foam, and glue (typically a combination of NBCA and ethiodized oil) (5,8,45). In 2013, Zhu et al (8) published their experience with percutaneous TSA PV catheterization for embolization of varices in 46 patients with PV complications, mostly tumor occlusion, preventing other modes of access. The technical success rate was 96%, and there were three incidents of procedure-related major bleeding (6.5%). Variceal embolization was combined with placement of a PV stent in five patients and TIPS in three patients. In a 2001 study, Gong et al (5) further described TSA used for esophageal and gastric fundal embolization in 18 patients with hepatocellular carcinoma. Technical success was 89%. Further details are provided in Tables 1 and 2. An example of TSA embolization of gastric varices is provided in Figure 4.

Pediatric Interventions

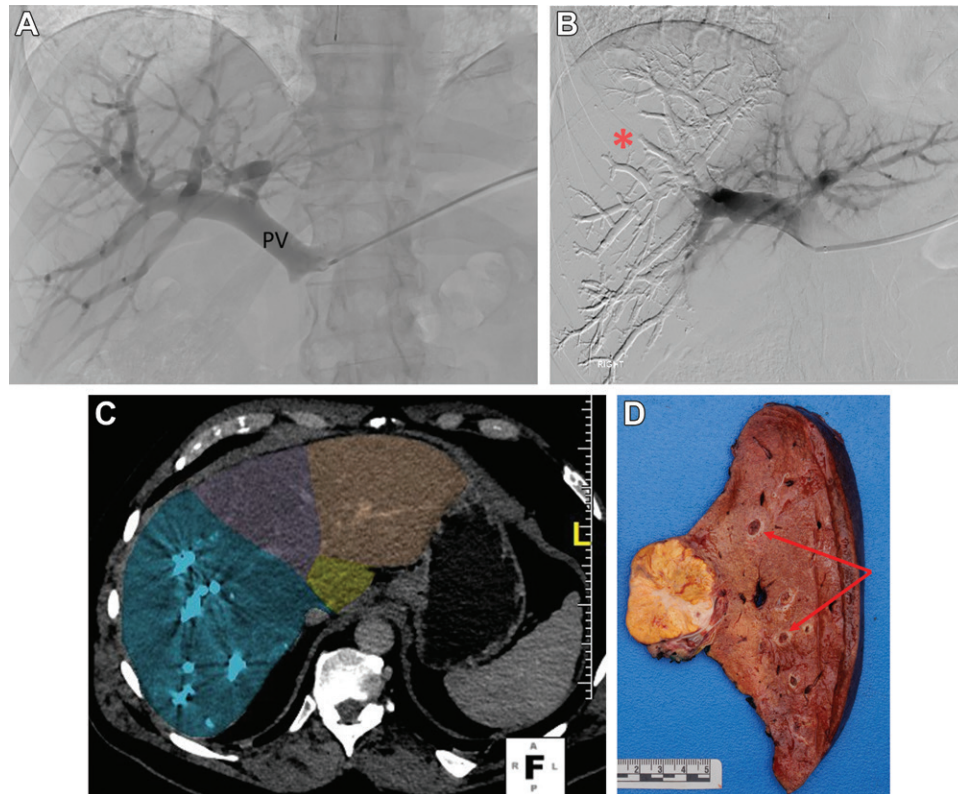
TSA of the PV in the pediatric population has been demonstrated for many of the same indications for which it is used in the adult population (46,47). One additional benefit in the pediatric population is the added value of splenoportography when the evaluation of the portomesenteric vasculature with noninvasive imaging is limited by patient size (48). The safety of TSA PV interventions in pediatric patients has varied between case series. In 2017, Pimpalwar et al (47) published their experience with 44 TSA interventions and reported intraperitoneal hemorrhage and the need for blood transfusion at rates higher than those of similar adult series: up to 27%. In a retrospective review published the following year, Monroe et al (48) reported their experience with 26 pediatric patients undergoing PV interventions with TSA and described intra-abdominal bleeding rates more in line with those in the adult series, with 7.7% of patients demonstrating intraperitoneal hemorrhage and only 3.8% requiring transfusion. Both articles (47,48) reported perfect technical success rates.

Closure

TSA Tract Embolization

Authors of multiple studies (4–6,8–11,20,21,49,50) have described various techniques of TSA

Figure 9. PVE in a 69-year-old man with a history of a 5-cm hepatocellular carcinoma in the right lobe, segment 6, who was planning to undergo right hepatectomy. Preoperative CT segmentation (not shown) of the liver demonstrated a future liver remnant of 25%. PVE was subsequently performed with a TSA approach, and US-guided access was obtained with a 21-gauge needle, followed by placement of a 6F sheath and a 5F catheter. (A) DSA image shows PVE performed through a 2.8F microcatheter that was used to subselect multiple branches of the right PV, followed by injection of a 6:1 dilution of ethiodized oil and N-butyl cyanoacrylate (NBCA) glue while retracting the catheter. (B) Postembolization DSA image shows successful embolization of the right PV tree (*). (C) Postembolization CT segmentation image acquired 1 month later shows that the future liver remnant (segments 1–4, yellow, purple, and orange areas) has increased to 38%. (D) Photograph shows the liver specimen after right hepatectomy, with PV thrombosis (arrows). Hepatocyte atrophy and PV inflammation were noted in the pathologic report.



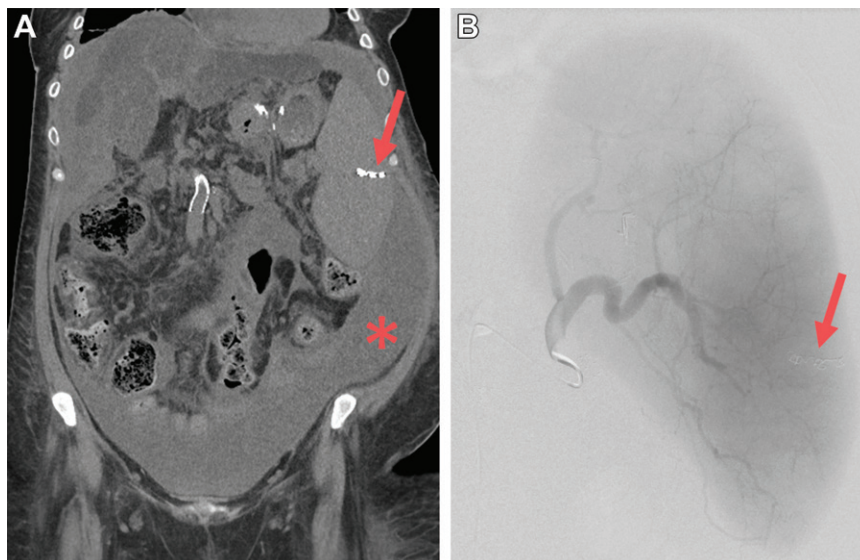
tract embolization for the purposes of reducing postprocedural hemorrhagic complications after completion of an endovascular PV intervention. Parenchymal tract embolization materials include vascular plugs, coils, gelatin sponges, mixtures of NBCA and ethiodized oil, or combinations of these materials. While these embolic materials are commonly used, the use of microfibrillar collagen or vascular closure devices (Angioseal; Terumo) are also being explored as alternative percutaneous tract embolic materials in transhepatic access and TSA. Rigual et al (51) examined the use of microfibrillar collagen in the embolization of the transsplenic tract. While this was found to display similar efficacy to that of coils or NBCA when used alone or in combination with other embolic agents, the presence of complications after its use indicates the need for further studies. In addition, Pescatori et al (52) demonstrated successful embolization with transhepatic access through simultaneous placement of two parallel vascular

closure devices with minimal to no postprocedural complications. Although these results demonstrate the potential for these embolic agents in TSA tract embolization, future studies of these materials are necessary to determine their safety and efficacy. A summary of commonly used embolization techniques is provided in Table 1, and examples of access tract embolization are displayed in Figure 5.

Postprocedure Care

Although no standardized protocol exists for postprocedural monitoring in patients undergoing TSA, the timing of bed rest, hospital admission, and administration of antibiotics or anticoagulation therapy is largely based on the intended procedure for which TSA was used. Length of bed rest is dependent on operator preference, with Liang et al (49) reporting 1 hour. Of the prospective and retrospective studies provided in this review (5,6,8,9,11,20,21,49), all have described patients who were admitted to either

Figure 10. Hemoperitoneum after TSA for placement of stents in the PV and superior mesenteric vein in a 45-year-old woman with PVT who developed abdominal pain, hypotension, abdominal swelling, and pain on postprocedure day 1. **(A)** Coronal noncontrast CT image shows high-attenuation free fluid in the abdomen, which is consistent with hemoperitoneum (*), and coil embolization of the TSA tract in the spleen (arrow). **(B)** Splenic artery angiogram shows no active extravasation, specifically at the site of access (arrow). The patient required resuscitation in the intensive care unit because of hemorrhagic shock and respiratory failure; however, she ultimately recovered and was discharged 11 days later.



an internal medicine service or an intensive care unit for 1 day, with vital signs and complete blood count obtained at various intervals. Combinations of US, CT, and venography have been described for imaging surveillance.

Complications

Reported complication rates and adverse events related to TSA in adults from prospective and retrospective studies are provided in Table 2. Commonly reported complications of TSA have largely been related to bleeding events, with an overall low rate of clinically significant hemorrhages requiring transfusion. Although most included studies (8,9,11,49) reported transfusion requirements in 0%–10% of patients, Gong et al (5) reported a higher rate of transfusion (28%) in patients undergoing TSA embolization of esophageal and gastric varices (5,6,8,9,11,20,21,49). In addition, Zhu et al (8) reported major bleeding complications in three of 46 patients after TSA procedures, including intraperitoneal hemorrhage and splenic subcapsular hemorrhage. Similarly, hemoperitoneum and intrasplenic hematoma have also been reported, which may indicate the need for further technical experience and improvements to prevent procedural bleeding complications. Contrast-enhanced CT is recommended to assess for vascular or hemorrhagic complications in patients with concerning clinical signs or symptoms. High-attenuation free fluid may be seen throughout the abdomen in patients with hemoperitoneum from TSA. In those with perisplenic hematoma related to TSA or perihepatic hematoma during TIPS needle access, organized fluid collections of varying attenuation may be identified and are related to evolving blood products. An example of hemoperitoneum

after TSA is provided in Figure 10, and small perisplenic or perihepatic hemorrhage after TSA placement of a TIPS is shown in Figure 7F.

To our knowledge, no current studies have been performed to assess the optimal closure techniques for TSA or the relationship between bleeding complications and the type of embolic used. The included studies (4–6,8–11,20,21,49,50) used primarily coils, glue, and gelfoam; however, we prefer to use Amplatzer vascular plugs, which have also been described in a single case by Dollinger et al (50). Furthermore, wide variability is reported in the types of access devices, needle gauges, and sheath sizes used for TSA and PV intervention. To our knowledge, it remains unclear if bleeding complications are related to these variables. Pain after TSA is common and can be managed with conservative pain control (11). Additional described complications of TSA include pneumothorax and pleural effusion, which are thought to be related to high percutaneous access (5,8,49). Other uncommon major adverse events such as SV dissection and acute SV rupture have also been observed (11,20,53). Although multiple factors may be responsible for such complications, the improvement of maneuvering techniques or the avoidance of aggressive sheath access may be crucial in preventing these events. Nontarget embolization of tract embolization materials has also been described; specifically, NBCA glue material migration in the SV. A summary of important technical tips to prevent pitfalls and complications related to TSA is provided in Table 3.

Contraindications

Contraindications of TSA are primarily related to bleeding risk and vascular patency. The

Table 3: Tips to Avoid Pitfalls with PV TSA

Ensure SV patency with preprocedural (CT or MRI) and periprocedural (US) imaging.
Perform routine laboratory studies including complete blood count and prothrombin time/international normalized ratio test, and make corrections for any results beyond Society of Interventional Radiology guidelines or institutional thresholds for procedures with a high bleeding risk.
Understand steps and techniques of TSA and subsequent intended TSA intervention(s), and have a plan for tract embolization.
Select a central intraparenchymal SV branch for access under US guidance. Be cognizant of needle trajectory to avoid transgression of the pleura superiorly.
Avoid multiple punctures to the spleen to reduce risk of hemorrhagic complications; do not enter the splenic capsule until the target vein and access needle are aligned.
Closure of the TSA tract is commonly performed. Described methods of embolization include the use of coils, NBCA-ethiodized oil, vascular plugs, gelatin sponges, microfibrillar collagen, or a combination of these agents.
Be prepared for postprocedural care. Have a plan for clinical and imaging follow-up of possible complications pertaining to TSA and the PV intervention performed.

authors' institution follows the 2019 Society of Interventional Radiology Consensus guidelines (54), which provide suggested laboratory value thresholds for performing a procedure. Before TSA is acquired in patients with normal liver function, the international normalized ratio should be corrected to within 1.5–1.8 or less, with platelet transfusion at levels lower than $50 \times 10^9/L$. In patients with chronic liver disease, suggested cutoffs are an international normalized ratio of less than 2.5, a platelet count greater than $30 \times 10^9/L$, and fibrinogen level greater than 100 mg/dL (1.00 g/L). However, more stringent values may be determined on the basis of operator preference. Viscoelastic testing, similar to thromboelastography, complements standard anticoagulation tests and may help providers with their decision whether to proceed with the procedure by better predicting bleeding risk (55). From a technical standpoint, SV and SV–superior mesenteric vein confluence patency are necessary for TSA, as are percutaneously accessible intraparenchymal SV branches. These parameters should be assessed at preprocedural cross-sectional imaging and periprocedural US, as previously discussed.

Conclusion

TSA is increasingly being used for PV interventions, for recanalization in chronic PV occlusion, stent placement in PV stenosis, preoperative PVE, complex TIPS placement procedures, and variceal embolization. For clinical scenarios in which a traditional transhepatic approach is not feasible in patients who require PV intervention, TSA has been shown to be a generally safe technique with high technical success rates. Bleeding complications remain the primary adverse event in TSA interventions; further research is needed to clarify if an ideal method of access and closure exists that may minimize postprocedural hemorrhage rates.

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